Automated Industrial Environmental Compliance Monitoring System

Using Multi-Temporal Sentinel-2 Imagery, KGIS and Machine Learning

Under the guidance of Dr. Sunil C K, Assistant Professor

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The GitHub repository is available here.

Problem Statement :

Manual environmental inspections of industrial zones are infrequent and hard to scale

Karnataka SPCB requires factories to maintain minimum green cover — compliance rarely verified

Need: Automated system using satellite imagery + ML to detect violations continuously

Statement 3 Requirements:

- Utilize KGIS boundary data to identify factory premises.
- Analyze satellite imagery over time — detect green cover loss & unauthorized construction.
- Apply ML models for vegetation detection and land-use change classification.
- Generate compliance reports with geo-coordinates, temporal analysis, visual proof.
- Enable scalable, continuous monitoring for pollution control boards.

Objectives :

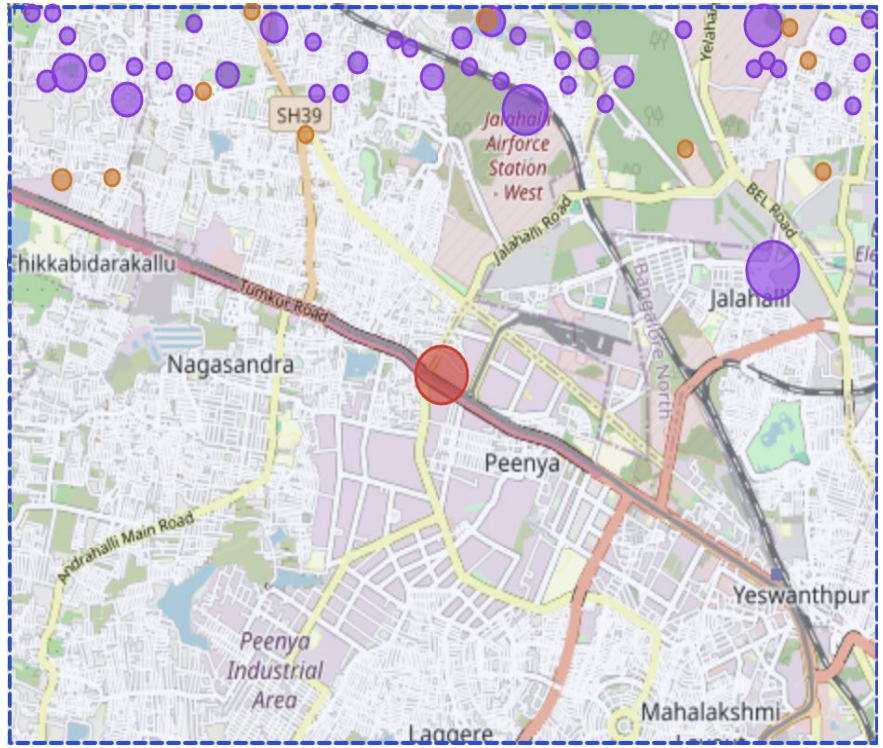
1. Build an end-to-end automated pipeline from data acquisition to compliance reporting.
2. Integrate KGIS taluk boundary data for precise industrial zone delineation.
3. Download and preprocess multi-temporal Sentinel-2 imagery (2020 vs 2024).
4. Compute spectral indices (NDVI, NDBI, NBI) for vegetation and built-up analysis.
5. Train Random Forest classifier for 6-class land cover classification.
6. Perform bitemporal change detection to quantify green cover loss.
7. Extract geo-tagged violation zones with coordinates and area estimates.
8. Generate HTML compliance reports and an interactive web dashboard.

Study Area — Peenya Industrial Area

Location: North-west Bengaluru, Karnataka.
KGIS Taluk: Bangalore-North (Code 2001).
Coordinates: 13.01–13.07°N, 77.48–77.56°E
Area: ~40 km²

Characteristics:

- One of the largest industrial estates in South Asia.
- 8,000+ registered manufacturing, electronics, and heavy industry units.
- Significant land-use transition due to metropolitan expansion.
- Dense mix of factories with mandated green belt requirements.



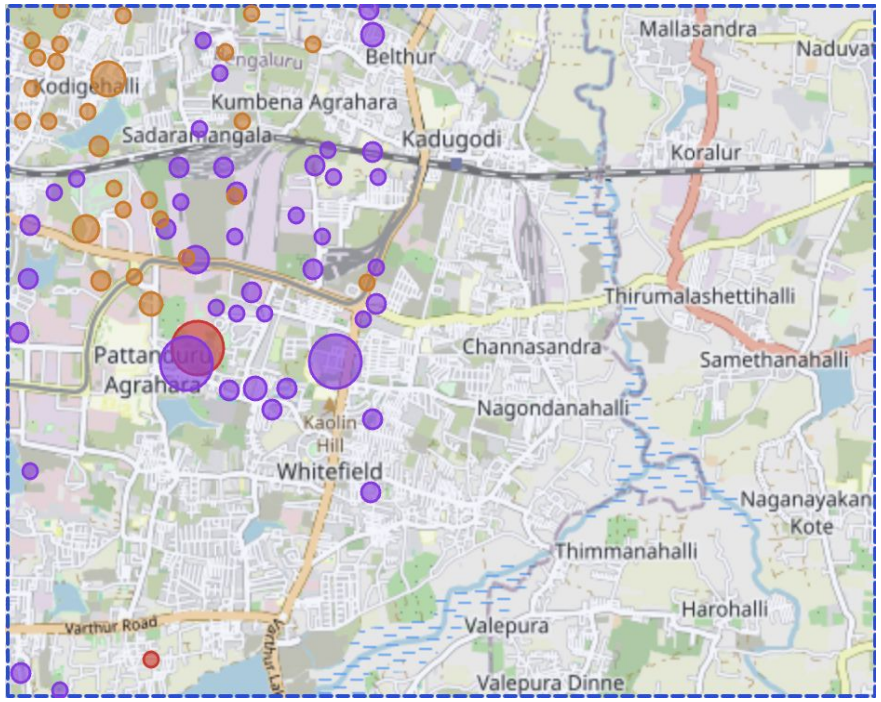
Peenya Industrial Areas

Study Area — Whitefield Industrial Area

Location: East Bengaluru, Karnataka
KGIS Taluk: Bangalore-East (Code 2004)
Coordinates: 12.95–13.01°N, 77.72–77.80°E
Area: ~25 km²

Characteristics:

- Rapidly urbanizing IT corridor (ITPL, major tech parks).
- Mixed industrial and residential development.
- Heavy construction activity driving green cover loss.
- Different industrial profile from Peenya — enables comparison.



Whitefield Industrial Area

Data Source 1 — KGIS (Karnataka GIS)

Source: kgis.ksrsac.in (KSRSAC portal)

What it provides:

- Administrative boundary shapefiles (taluk, district, ward level).
- Downloaded: Taluk.shp with all 226 Karnataka taluks.
- Filtered by: Code 2001 (Bangalore-North) and 2004 (Bangalore-East).
- Location Details REST API for ward/zone metadata per violation.

Purpose in project:

- Delineates industrial zone boundaries — tells system WHERE to analyze.
- Masks satellite imagery to only include pixels inside industrial premises.

Does NOT provide: satellite imagery, vegetation data, or temporal information.

Data Source 2 — Sentinel-2 L2A Satellite Imagery

Source: ESA Copernicus via Microsoft Planetary Computer (STAC API)

Free, open-access multispectral satellite imagery

Specifications:

- Resolution: 10m (B02–B08), 20m resampled to 10m (B11–B12).
- Bands: Blue, Green, Red, NIR, SWIR1, SWIR2 + SCL cloud mask.
- Time periods: March 2020 (baseline) vs March 2024 (recent).
- Cloud cover: <0.2% (T1), <0.01% (T2) — dry season selection.

Scale:

- 4 satellite scenes across 28 GeoTIFF files.
- ~11.6 million multispectral pixels total.
- 582,066 training + 249,458 test samples for ML.

System Architecture — Stages 1–3

Stage 1: Data Acquisition

- KGIS shapefile download from KSRSAC portal.
- Sentinel-2 STAC API query with cloud cover filter (<15%).
- Windowed GeoTIFF read (downloads only AOI, not full 100km tile).

Stage 2: Preprocessing

- SCL cloud masking (removes cloud, shadow, cirrus pixels).
- 20m → 10m bilinear resampling (B11, B12).
- KGIS boundary rasterization and spatial masking.
- 7-band GeoTIFF stacking [B02, B03, B04, B08, B11, B12, Mask].

Stage 3: Spectral Index Computation

- NDVI: vegetation health quantification.
- NDBI: built-up area detection.
- NBI: new construction identification.

System Architecture — Stages 4–6

Stage 4: ML Classification

- Random Forest (100 trees) trained on 6 raw spectral bands.
- K-Means (k=5) unsupervised clustering as baseline.
- 70/30 stratified train-test split with validation metrics.

Stage 5: Change Detection

- Bitemporal NDVI/NDBI differencing (2024 – 2020).
- Morphological closing to merge scattered detections.
- Connected component labeling for violation zone extraction.

Stage 6: Report Generation

- HTML compliance reports with risk assessment.
- Annotated satellite imagery with violation markers.
- Geo-coordinate export (lat/lon for each violation).
- Interactive web dashboard with Leaflet.js maps.

Preprocessing — Cloud Masking & Resampling

Cloud Masking (SCL Band)

- Scene Classification Layer identifies pixel types.
- Masked: Cloud shadow (3), Cloud medium (8), Cloud high (9), Cirrus (10).
- Valid: Vegetation (4), Not-vegetated (5), Water (6), Unclassified (7).
- Result: 99.8% valid pixel coverage across all 4 acquisitions.

Band Resampling

- B11 (SWIR1) and B12 (SWIR2) at 20m native resolution.
- Resampled to 10m using bilinear interpolation (scipy.ndimage.zoom).

KGIS Boundary Masking

- Taluk polygon reprojected from WGS84 to UTM Zone 43N.
- Rasterized onto satellite grid — pixels outside boundary = 0.
- Combined mask: pixel must be cloud-free AND inside KGIS boundary.

Spectral Indices

NDVI — Normalized Difference Vegetation Index

- Formula: (NIR – Red) / (NIR + Red).
- Range: [−1, +1] — higher values = more vegetation.
- Used for: Green cover quantification and vegetation loss detection.

NDBI — Normalized Difference Built-up Index

- Formula: (SWIR1 – NIR) / (SWIR1 + NIR).
- Range: [−1, +1] — higher values = more concrete/built-up.
- Used for: Unauthorized construction detection.

NBI — New Built-up Index

- Formula: (Red × SWIR1) / NIR.
- Higher values = bare soil or fresh construction.

Change Detection Methodology

Bitemporal Differencing

- $\Delta NDVI = NDVI_2024 - NDVI_2020 \rightarrow$ negative = vegetation lost.
- $\Delta NDBI = NDBI_2024 - NDBI_2020 \rightarrow$ positive = new construction.

Classification Thresholds

- Class 1 — Vegetation Loss: $\Delta NDVI < -0.15$.
- Class 2 — New Construction: $\Delta NDBI > +0.10$.
- Class 3 — Both conditions met simultaneously.
- Class 0 — No significant change.

Risk Assessment

- HIGH: green cover change < −30%.
- MEDIUM: green cover change between −15% and −30%.
- LOW: green cover change > −15%.

ML Classification — Random Forest

Algorithm: Random Forest (scikit-learn)

- 100 decision trees, max_depth=15, balanced class weights.
- min_samples_split=10, min_samples_leaf=5, n_jobs=-1.

Features (input): 6 raw spectral bands

- B02 (Blue), B03 (Green), B04 (Red), B08 (NIR), B11 (SWIR1), B12 (SWIR2).

Labels (output): 6 land cover classes

- Dense Vegetation, Sparse Vegetation, Built-up, Bare Soil, Water, Other.

Training setup:

- Each 10m×10m pixel = one sample with 6 reflectance values.
- 70% train / 30% test, stratified by class.
- K-Means (k=5) as unsupervised baseline for comparison.

Non-Circular Validation — Key Design Decision

The Problem:

- Training labels are derived from NDVI/NDBI thresholds.
- If RF trains on NDVI/NDBI as features → trivially memorizes thresholds.
- Would get meaningless 100% accuracy.

Our Solution:

- Labels generated from: NDVI and NDBI (spectral indices).
- RF trained on: Raw bands ONLY (B02, B03, B04, B08, B11, B12).
- Features and label sources are completely DISJOINT.

Why This Works:

- RF must learn spectral reflectance patterns from raw data.
- Cannot shortcut by reading NDVI/NDBI values directly.
- The 97–98% accuracy reflects genuine pattern recognition.

Violation Extraction & Geo-Tagging

Morphological Processing

- 5×5 structuring element, 3 iterations of closing.
- Merges scattered single-pixel detections within 50m into regions.

Connected Component Labeling

- SciPy labels contiguous violation clusters.
- Filters regions below 500 m^2 (noise removal).

Geo-Coordinate Conversion

- Centroid pixel → UTM Zone 43N → WGS84 (lat/lon) via PyProj.
- Each violation: latitude, longitude, type, area (hectares).

KGIS API Enrichment

- Queries KGIS Location Details REST API per violation.
- Adds ward name, zone name, town, district metadata.

Results — Change Detection

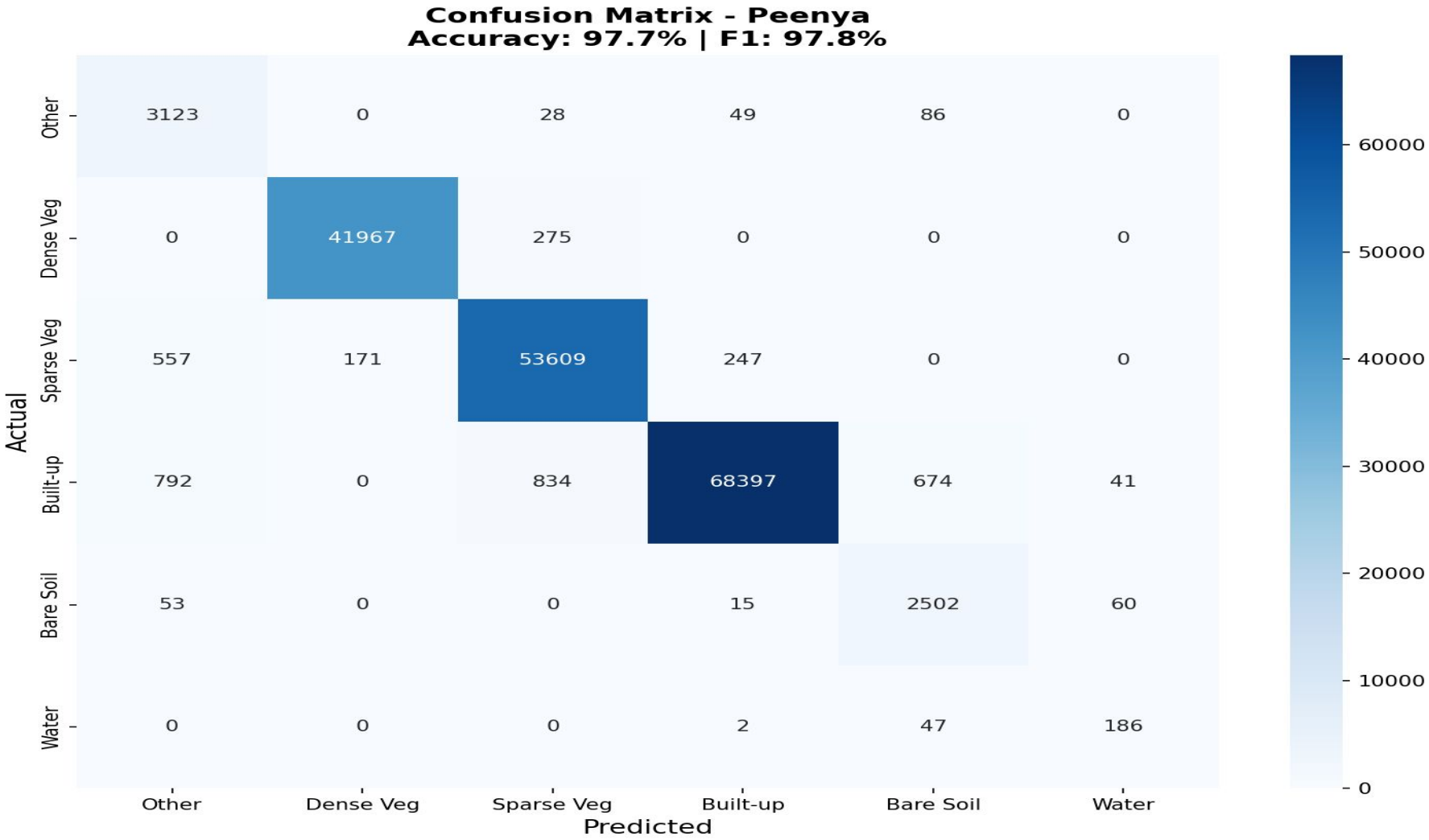
Metric	Peenya	Whitefield
Green Cover 2020	52.57%	71.25%
Green Cover 2024	28.51%	43.45%
Net Change	**−24.06%**	**−27.80%**
Vegetation Loss	27.23%	38.92%
New Construction	4.56%	17.72%
Violation Zones	54	72
Risk Level	MEDIUM	MEDIUM

Results — ML Model Performance

Metric	Peenya	Whitefield
Training Samples	405,333	176,733
Test Samples	173,715	75,743
Accuracy	**97.74%**	**98.70%**
Precision (weighted)	**98.06%**	**98.79%**
Recall (weighted)	**97.74%**	**98.70%**
F1-Score (weighted)	**97.83%**	**98.73%**

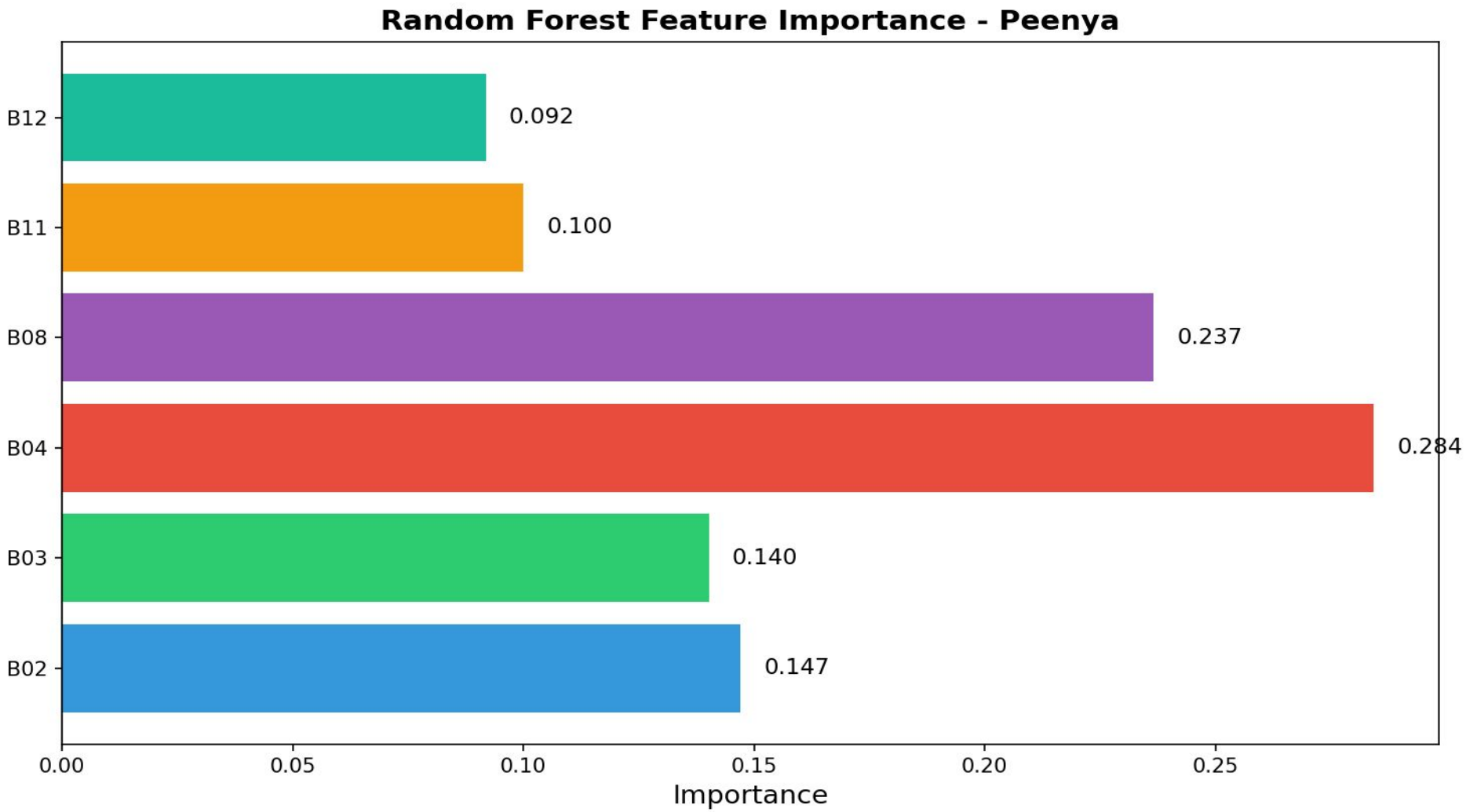
Non-circular validation: RF trained on raw bands only, labels from NDVI/NDBI thresholds

Results — Confusion Matrix (Peenya)



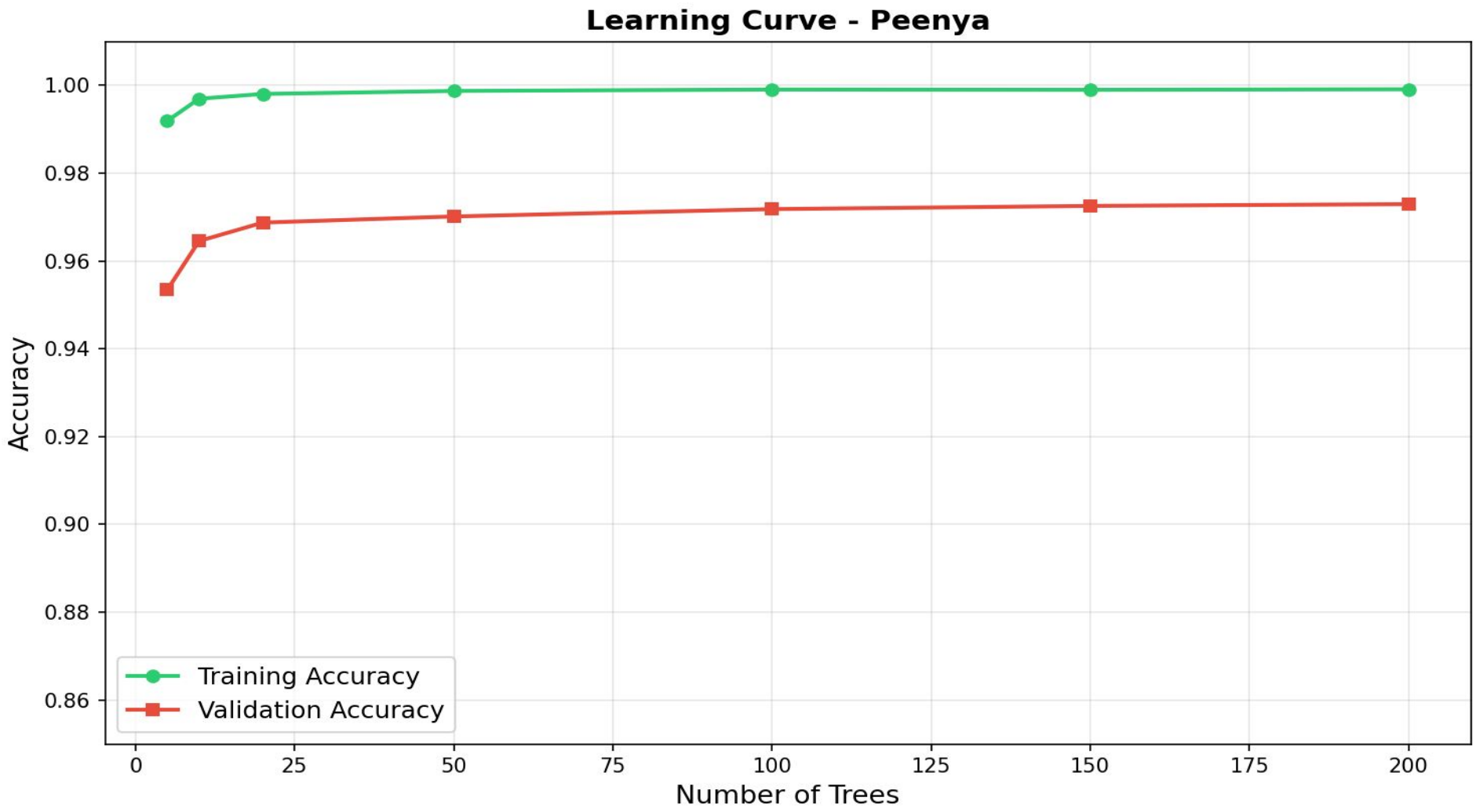
6×6 matrix on 173,715 test pixels — strong diagonal dominance across all classes

Results — Feature Importance



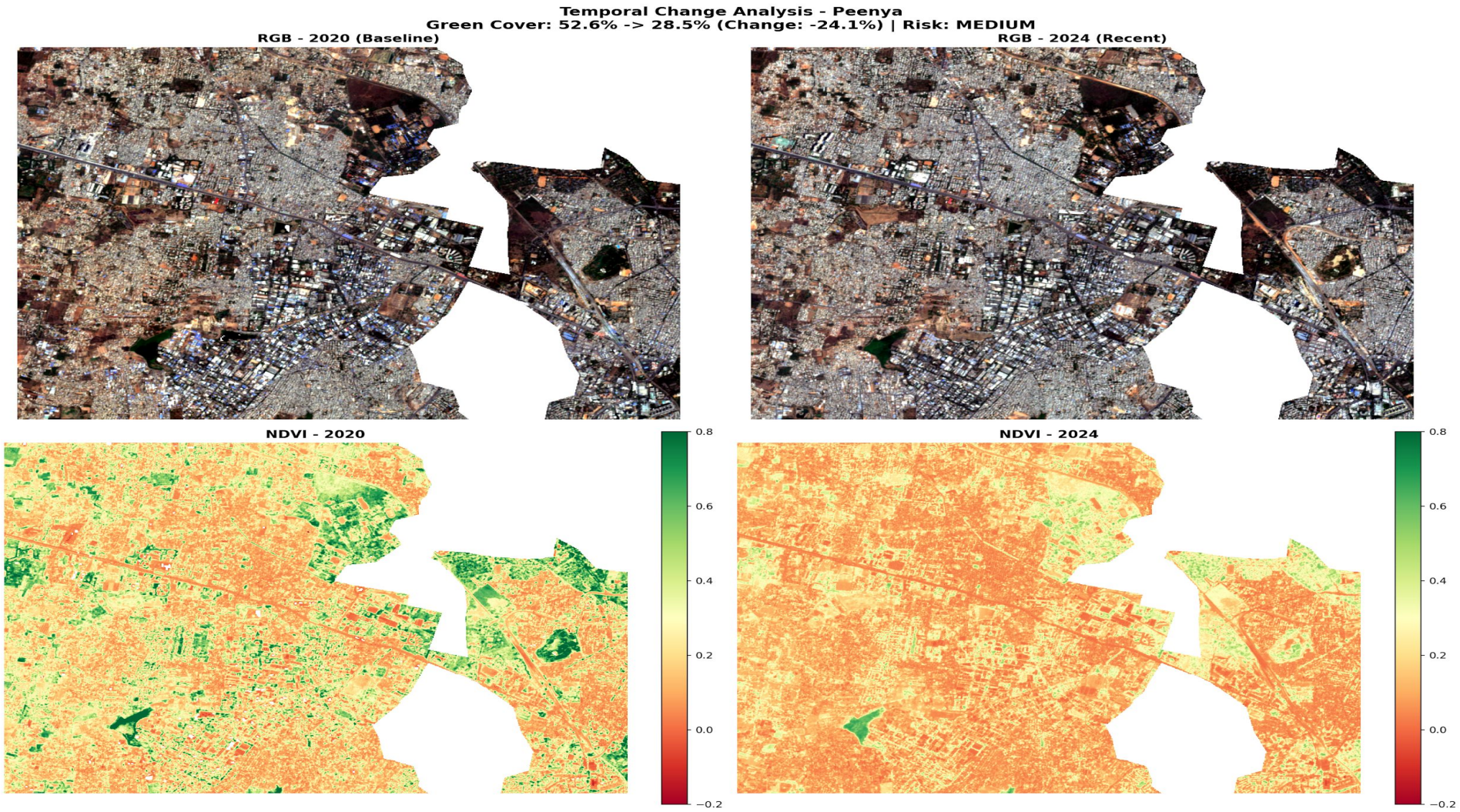
B04 (Red, 28.4%) and B08 (NIR, 23.7%) dominate — consistent with vegetation spectral theory

Results — Learning Curve



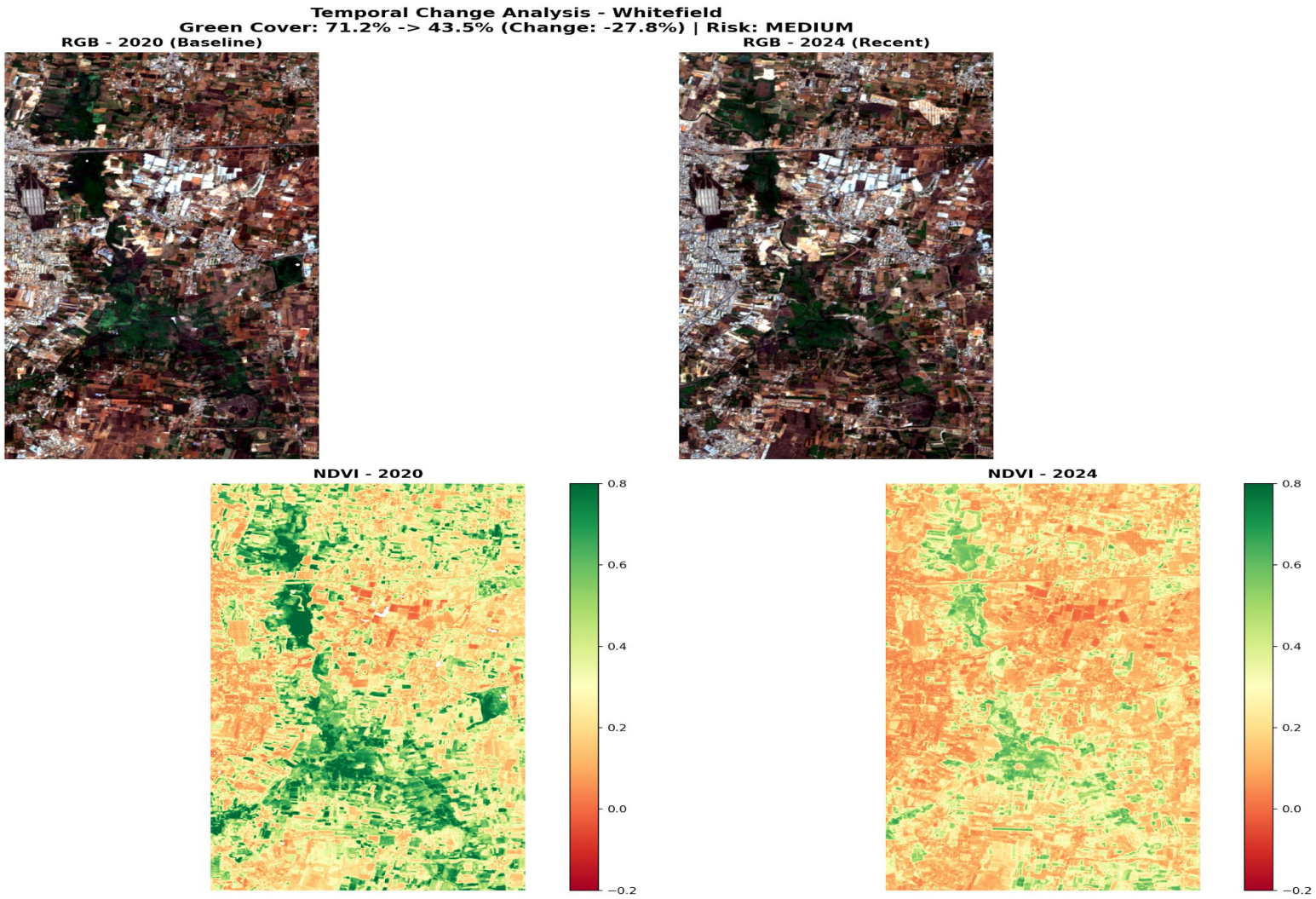
Accuracy stabilizes at ~50 trees, confirming adequate model capacity without overfitting

Results — Before/After Analysis (Peenya)



Top: RGB 2020 vs 2024 | Bottom: NDVI heatmaps showing vegetation density decline

Results — Before/After Analysis (Whitefield)

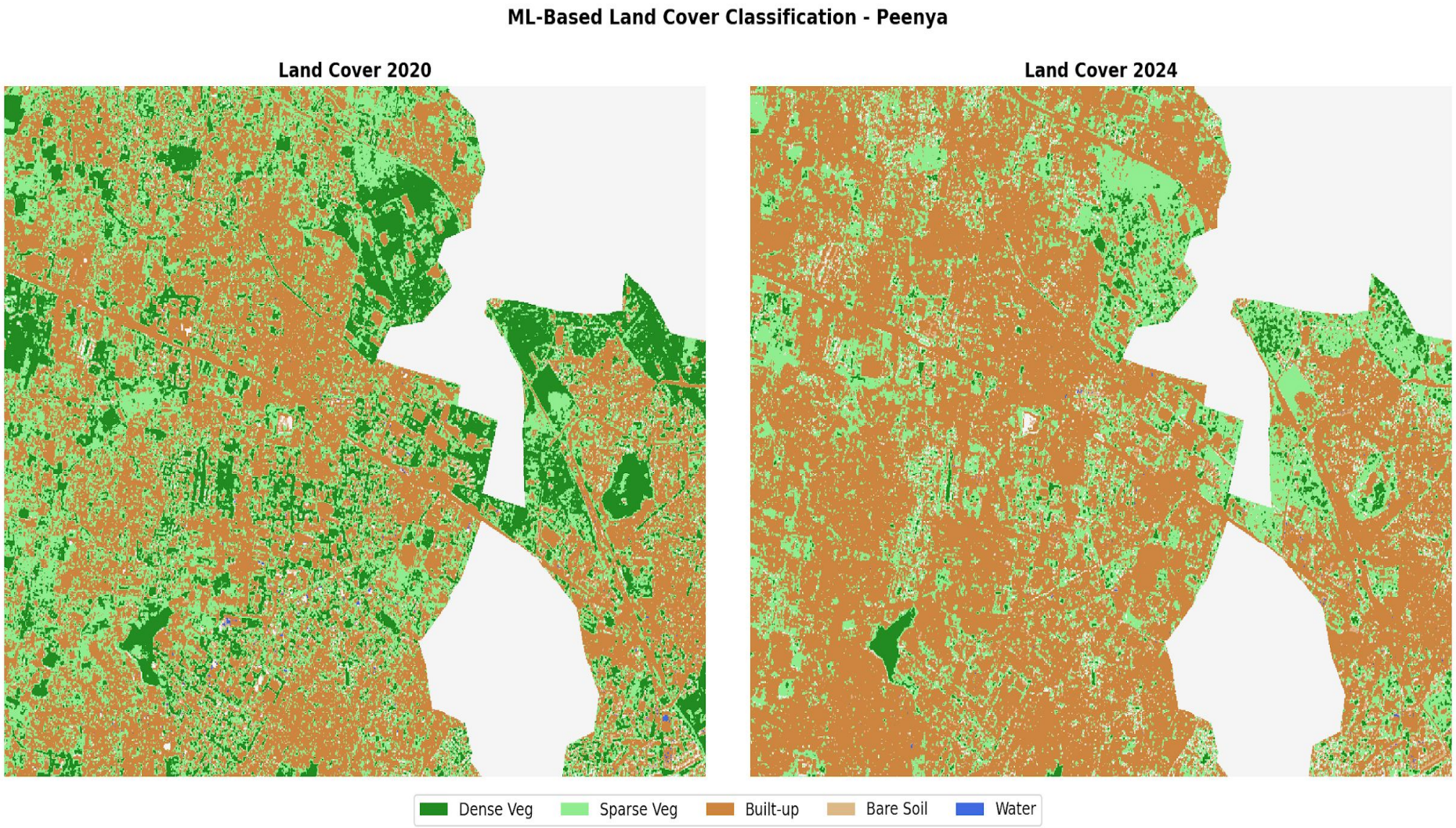


Severe vegetation loss in IT corridor — 71.25% → 43.45% green cover

Results — Annotated Violation Map

Red: vegetation loss | Orange: new construction | Pink: both — geo-tagged markers

Results — Land Cover Classification



ML-based land cover: 2020 vs 2024 — clear vegetation-to-built-up transition

Web Dashboard — Interactive Monitoring

Technology: Flask (Python) backend + HTML/JS/CSS frontend.

Features:

- Interactive Leaflet.js map with color-coded violation markers.
- Zone switching tabs (Peenya / Whitefield).
- 6 metric cards: Green Cover, Net Change, Veg Loss, Construction, Risk.
- Violations table with geo-coordinates and area in hectares.
- ML performance panel with feature importance bars.
- Tabbed satellite imagery viewer (7 image types).

Data Flow:

- Pipeline runs OFFLINE (once) → generates JSON + PNG files.
- Flask reads pre-computed files → serves to browser.
- Zero computation at runtime — pure file serving.

The deployed web dashboard is available here.

Scalability & Extensibility

Geographic Scalability

- Pipeline parameterized by bounding box coordinates.
- Add new zone = add one bbox line + re-run pipeline.
- KGIS shapefile has all 226 Karnataka taluks.
- Sentinel-2 freely available globally.

Temporal Scalability

- Sentinel-2 revisits every 5 days.
- Pipeline can be scheduled (cron) for continuous monitoring.

Modular Architecture

- Each stage is an independent Python module.
- Any module can be replaced or upgraded independently.
- preprocess → indices → change_detection → ml_classifier → visualize.

Data Pulling and Preprocessing

Data Pulling:

- KGIS: Taluk shapefile (18.9 MB) downloaded from kgis.ksrsac.in.
- Sentinel-2: STAC API query → windowed read (only AOI pixels, not full tile).
- Automated: python download_data.py handles everything programmatically.
- 28 GeoTIFF files downloaded across 4 scenes (2 zones × 2 time periods).

Preprocessing Pipeline:

- Cloud Masking: SCL band removes clouds/shadows → 99.8% valid pixels.
- Resampling: 20m bands (B11, B12) → 10m via bilinear interpolation.
- KGIS Masking: Taluk polygon rasterized → clips to industrial boundary.
- Band Stacking: 7-band GeoTIFF [B02, B03, B04, B08, B11, B12, Mask].
- Index Computation: NDVI, NDBI, NBI with NaN handling for masked pixels.

Data Scale:

- ~11.6 million multispectral pixels • 582,066 training + 249,458 test samples.

Technique, Logic & Data Handling

Techniques Used:

- Spectral Analysis: NDVI, NDBI, NBI indices from multispectral reflectance
- Supervised ML: Random Forest (100 trees, max_depth=15, balanced weights)
- Unsupervised ML: K-Means clustering (k=5) as comparison baseline
- Change Detection: Bitemporal differencing with adaptive thresholds
- Morphological Analysis: Closing + connected components for violation extraction

Logic — Non-Circular Validation:

- Labels from NDVI/NDBI thresholds, but RF trains on raw bands ONLY
- Prevents trivial memorization → 97–98% accuracy is genuine

Data Handling:

- NumPy arrays with NaN masking for cloud/boundary pixels
- Safe division (zero-handling) in all index computations
- Windowed reads avoid downloading 100km tiles (only AOI pixels)
- CRS reprojection (WGS84 ↔ UTM) handled automatically by PyProj

Response Time:

- Pipeline: ~5 min total (download → results)
- Dashboard: instant (pre-computed)

Accuracy and Validation

Train/Test Split:

- 70% training / 30% test, stratified by class, random_state=42
- Peenya: 405,333 train + 173,715 test samples
- Whitefield: 176,733 train + 75,743 test samples

Classification Metrics (Test Set):

- Peenya: Accuracy 97.74%, F1 97.83%, Precision 98.06%, Recall 97.74%
- Whitefield: Accuracy 98.70%, F1 98.73%, Precision 98.79%, Recall 98.70%

Validation Artifacts:

- Confusion Matrix: 6×6 heatmap — strong diagonal dominance
- Feature Importance: B04 (Red, 28.4%) and B08 (NIR, 23.7%) dominate
- Learning Curve: accuracy vs n_trees (5–200) — stabilizes at ~50 trees
- Per-class F1: Dense Veg 0.99, Sparse Veg 0.98, Built-up 0.98

Key Validation Design:

- Non-circular: features ≠ label source → accuracy reflects real learning
- Balanced class weights handle minority classes (Water, Bare Soil)

Analysis Summary — Key Findings

Green Cover Analysis:

- Peenya: 52.57% → 28.51% (−24.06%) — factory expansion driving loss
- Whitefield: 71.25% → 43.45% (−27.80%) — IT park + residential construction

Unauthorized Construction:

- Peenya: 4.56% new construction detected
- Whitefield: 17.72% new construction — 4x more than Peenya

Violation Detection:

- 126 total violation zones identified (54 Peenya + 72 Whitefield)
- Largest: 1,457 ha vegetation loss in Peenya (13.04°N, 77.52°E)
- Each violation has: lat/lon, type, area, KGIS ward metadata

ML Model Insights:

- B04 (Red) and B08 (NIR) are most important — these form NDVI
- Whitefield has higher accuracy (98.7%) due to more homogeneous landscape
- Learning curve shows no overfitting — performance stable after 50 trees

Compliance Report Generation

Auto-Generated HTML Reports (per zone):

- Risk assessment badge (LOW / MEDIUM / HIGH) with color coding
- Green cover metrics: 2020 vs 2024 with percentage change
- ML model performance table (accuracy, precision, recall, F1)
- Violation alerts table with geo-coordinates and area in hectares
- Methodology documentation (6-step pipeline summary)

Visual Evidence Generated:

- Before/After panels: RGB + NDVI comparison (2020 vs 2024)
- Annotated satellite imagery: violations overlaid on 2024 image
- Land cover classification maps: side-by-side 2020 vs 2024
- Confusion matrix heatmaps, feature importance, learning curves

Report Formats:

- HTML compliance reports: reports/compliance_report_{zone}.html
- LaTeX academic paper: IEEE conference format (report.tex → report.pdf)
- Interactive dashboard: Flask + Leaflet.js web application

Conclusion

1. Built end-to-end automated pipeline for industrial compliance monitoring

2. Integrated KGIS taluk boundary data for precise study area delineation

3. Analyzed two zones across 2020–2024 using Sentinel-2 satellite imagery

4. Achieved 97.7–98.7% classification accuracy with non-circular RF validation

5. Detected 126 geo-tagged violation zones (54 Peenya + 72 Whitefield)

6. Quantified green cover loss: −24.1% (Peenya) and −27.8% (Whitefield)

7. Generated automated compliance reports with visual evidence

Thank You